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AMOC

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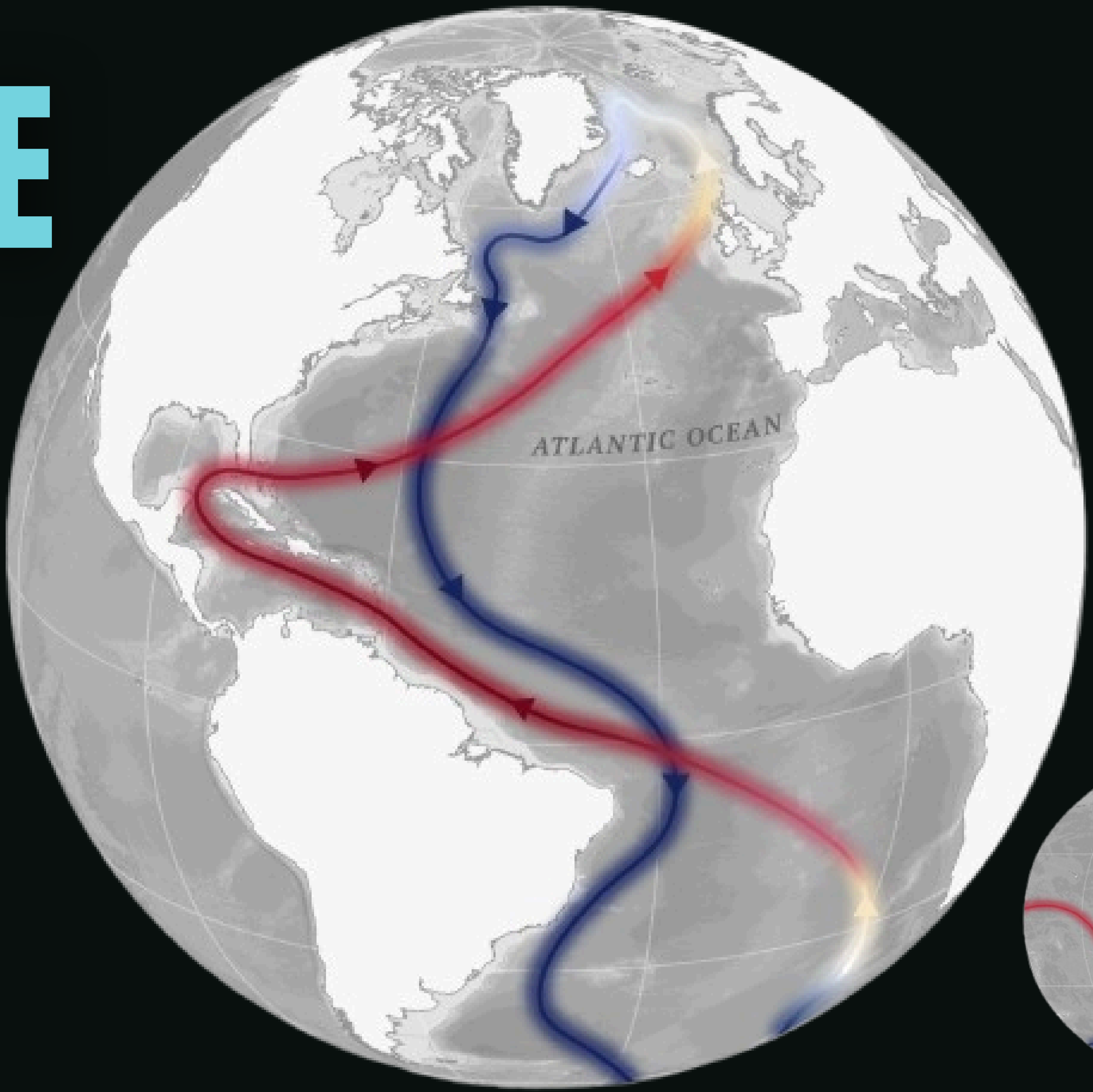
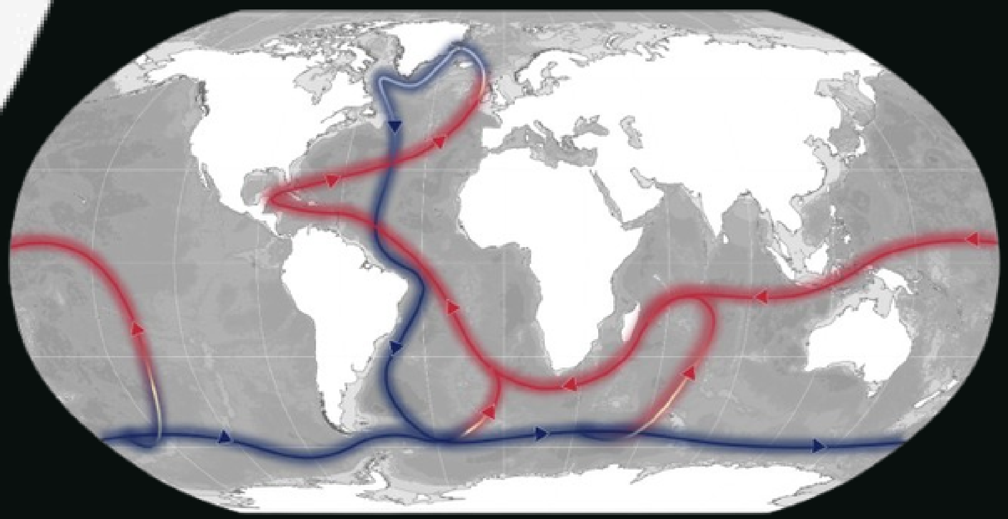
WHAT IS THE AMOC?

The Atlantic Meridional Overturning Circulation (AMOC) is a large system of ocean currents that redistributes heat throughout the Atlantic Ocean. Warm surface waters travel northwards from the tropics, whilst colder, denser waters return southwards at depth.

This circulation forms part of a larger global ocean system that transports heat, nutrients and carbon around the planet.



- Transfers warm tropical water towards Europe.
- Returns cold, deep water southwards.
- Forms part of the global ocean circulation system.
- Helps regulate climate, rainfall and marine ecosystems.
- “Ocean conveyor belt”



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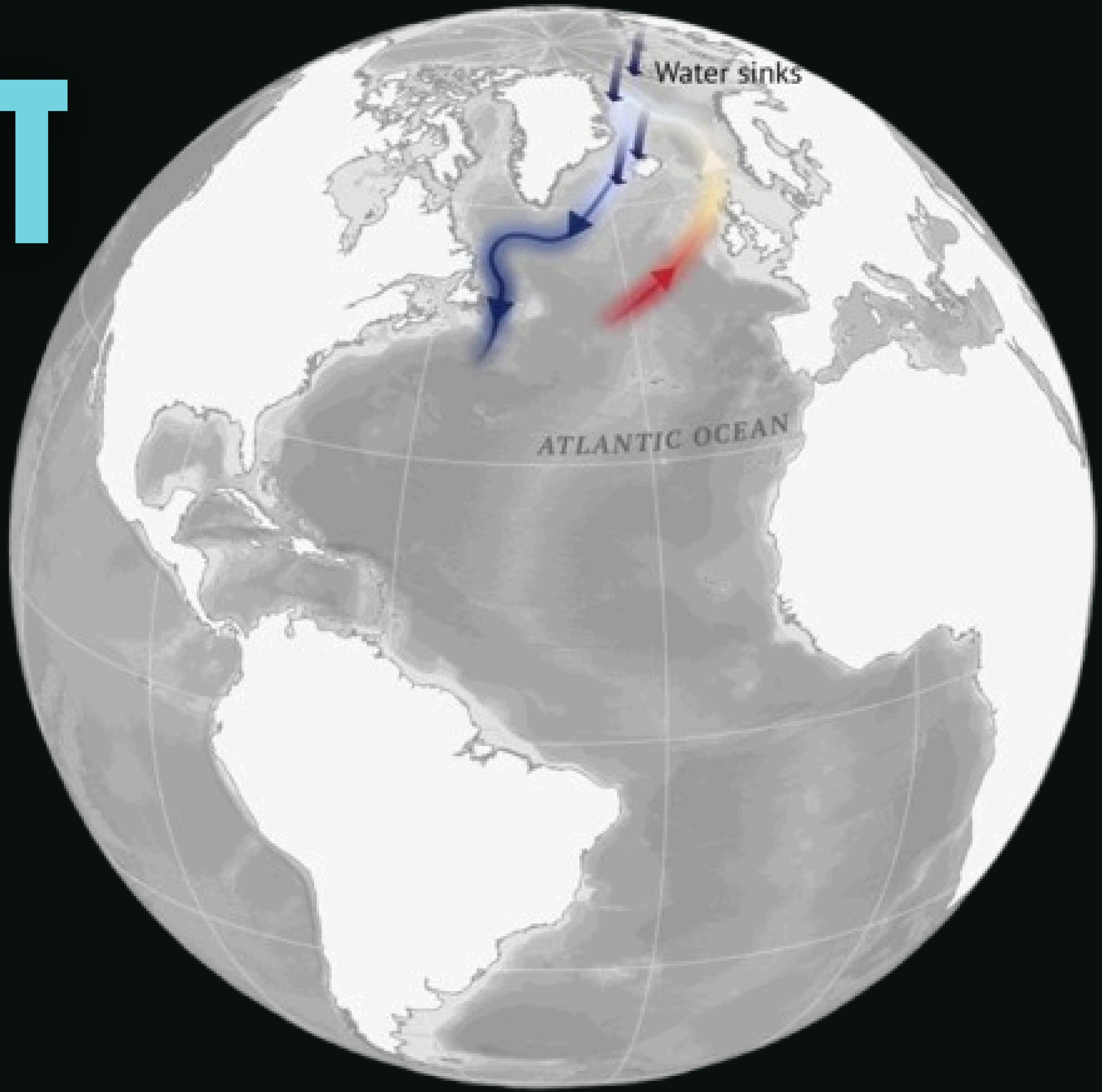
HOW DOES IT WORK?

In the high latitudes of the North Atlantic, warm surface water is cooled by the overlying atmosphere. As water evaporates and sea ice forms, salt is left behind in the ocean. As a result, the surface water sinks.



01. Dense water sinks near greenland

02. Deep water flows southwards

03. Southern Ocean upwelling

04. Warm water moves northwards

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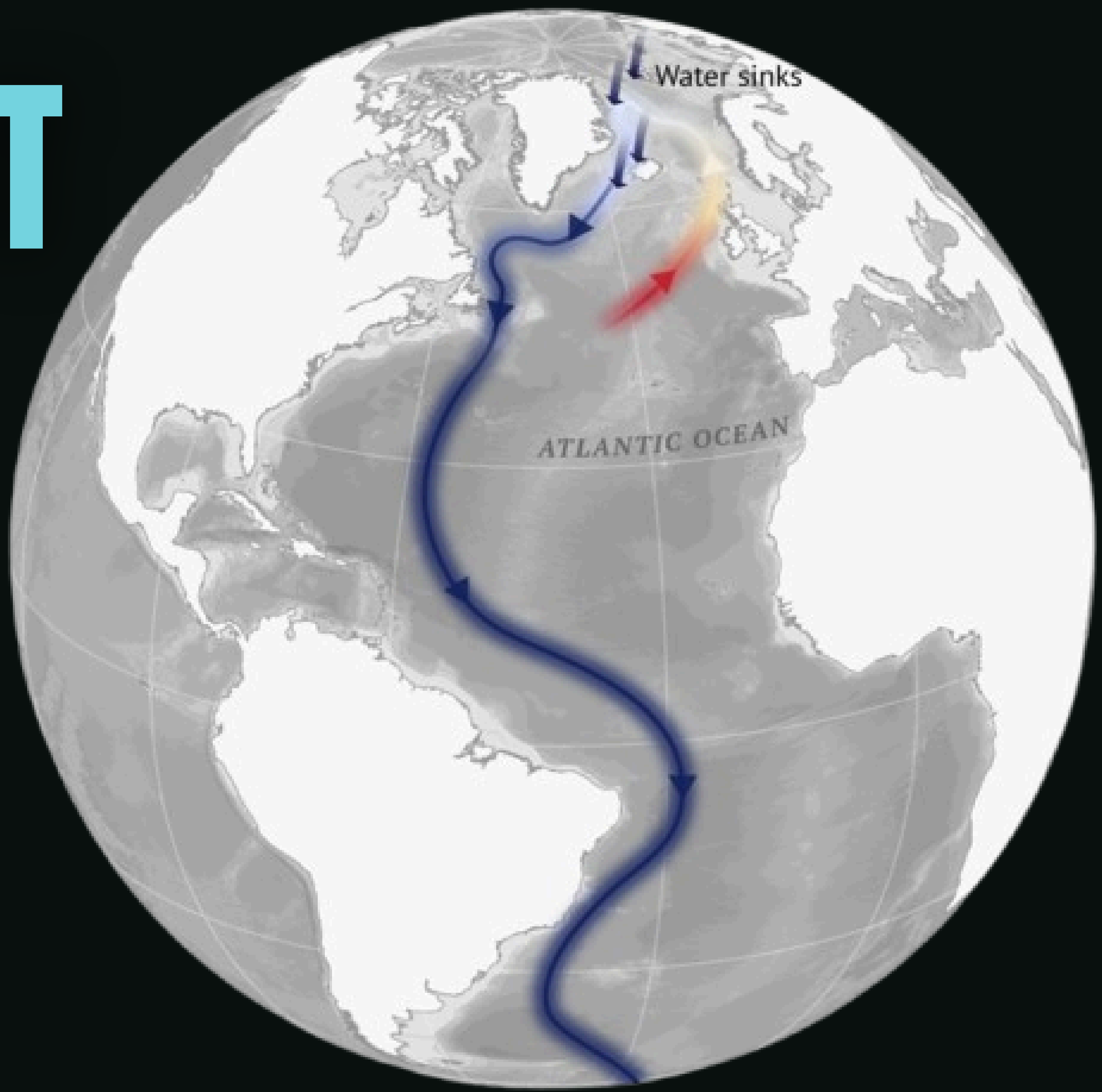
HOW DOES IT WORK?

The water then heads south, thousands of metres below the ocean's surface.



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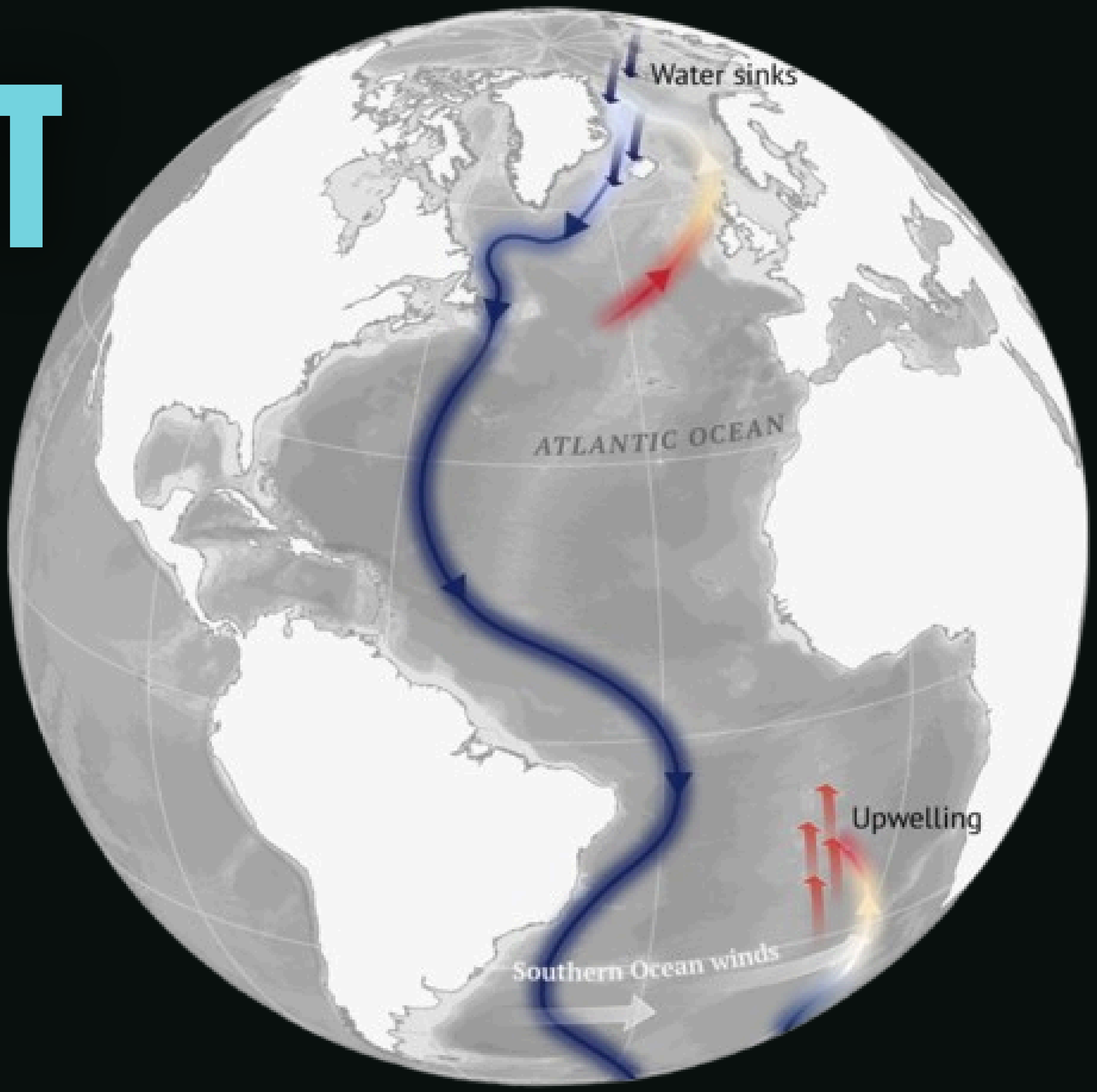

Outside of the Atlantic, this cold water is pulled back to the surface, assisted by ocean mixing and winds in the Southern Ocean. This is known as “upwelling”.

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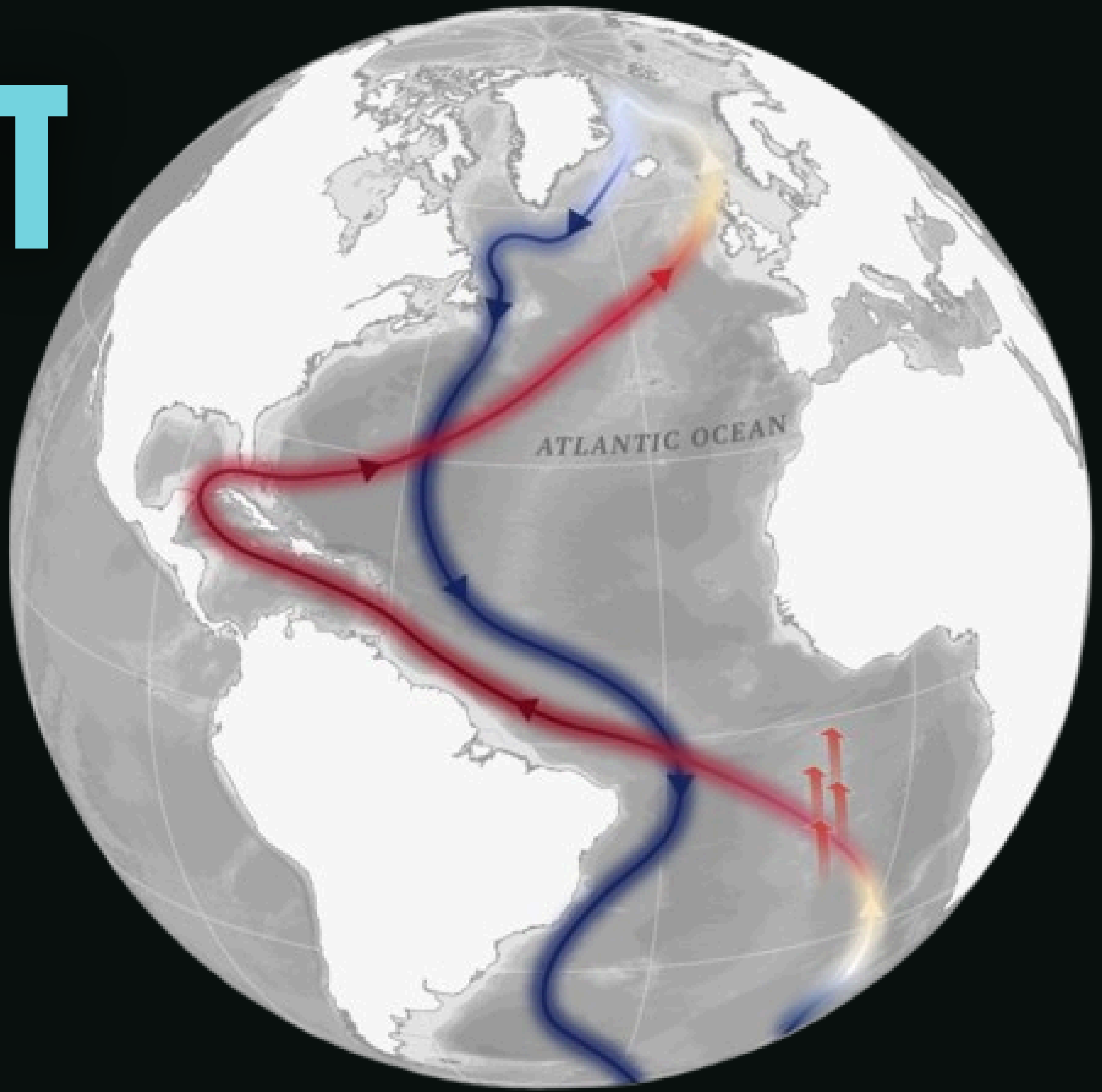

Upon returning to the North Atlantic, warm, salty water in the tropics is then pulled northwards towards Europe, closing the loop.

01. Dense water sinks near greenland

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WHY DO WE NEED IT?

The AMOC influences weather and climate patterns across the North Atlantic region and beyond. By transporting warm water northwards, it contributes to the relatively mild climate experienced in north-west Europe compared with other regions at similar latitudes.

The effects of the AMOC extend far beyond temperature alone, influencing rainfall patterns, storm tracks, marine ecosystems and sea level.

Example: London experiences much milder winters than regions of Canada located at similar latitudes, partly due to heat transported by the Atlantic Ocean.

01.

Helps moderate temperatures across Europe.

02.

Influences rainfall and drought patterns.

03.

Affects ocean ecosystems and fisheries.

04.

Contributes to global heat redistribution.

05.

Impacts regional sea level around the Atlantic basin.

WHAT IF IT SLOWS?

Observational studies and climate models suggest that a weakening AMOC could have significant impacts on regional and global climate. Whilst a complete collapse remains uncertain, even a substantial slowdown may alter weather patterns and ocean conditions.

Recent studies suggest that the AMOC may already be weaker than it was during the twentieth century, although the magnitude of this change remains an active area of research.

01.

Cooler temperatures across parts of Europe.

02.

Changes in rainfall distribution.

03.

Increased risk of droughts and flooding.

04.

Rising sea levels along some Atlantic coastlines.

05.

Disruption of marine ecosystems and fisheries.

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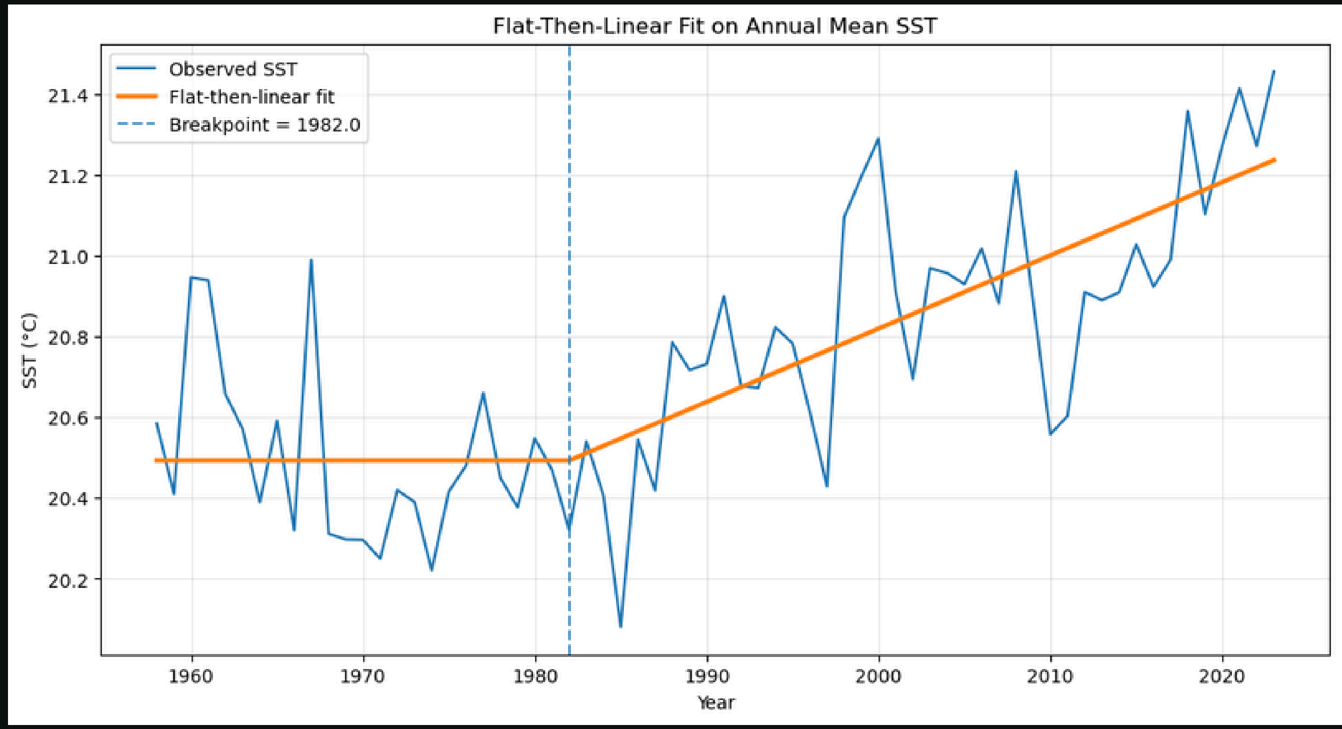

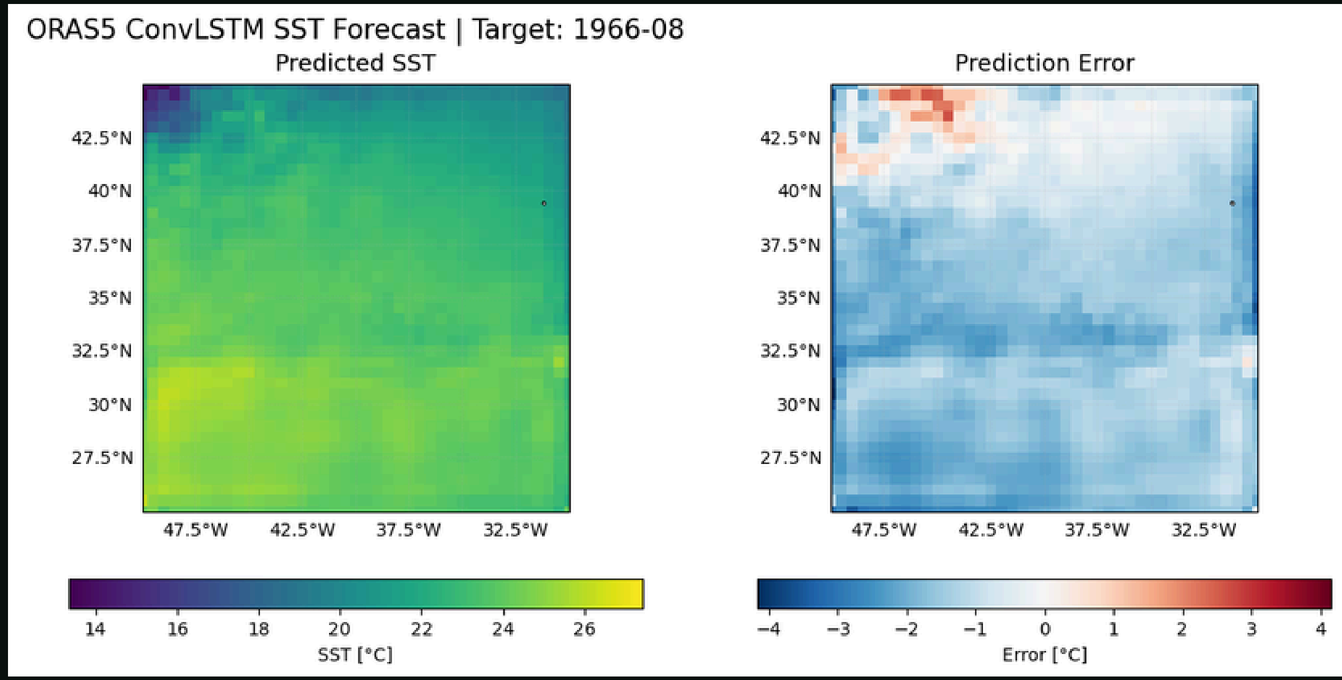

OUR QUESTION

Can artificial intelligence identify early warning signals of AMOC change?

Modern climate datasets contain vast amounts of information describing the Atlantic Ocean and atmosphere. Machine learning offers a new way to identify subtle patterns that may not be visible using traditional statistical approaches.

01. Inflection Year Study

- Both statistical and machine learning methods
- Finds the point at which things start to change

02. Anomaly Detection

- Machine learning is taught based on the ‘normal’ data
- Predicts what happens after
- Reconstruction error indicates change

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THE DATA

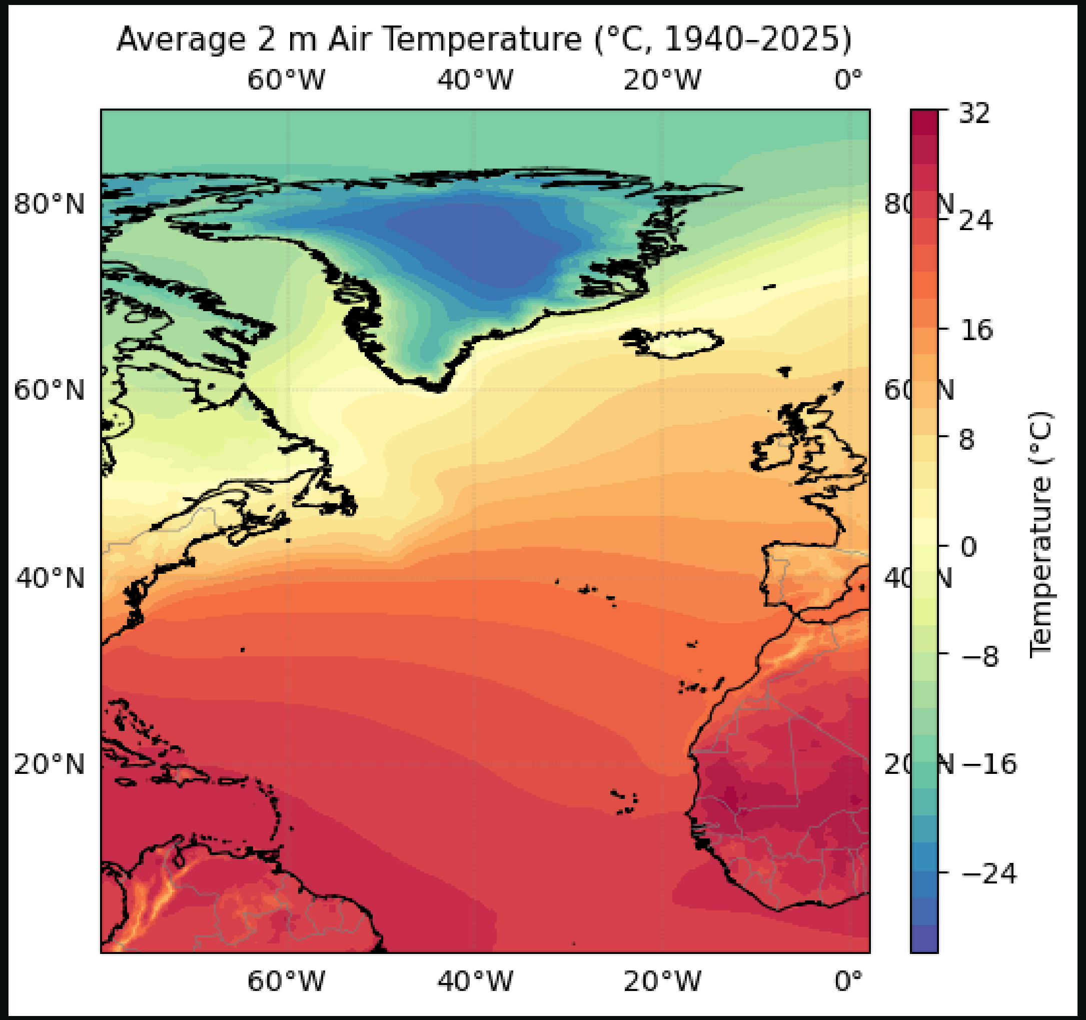
This project combines atmospheric and ocean reanalysis datasets from the Copernicus Climate Change Service. These datasets provide decades of observations describing the state of the Atlantic Ocean and surrounding atmosphere.

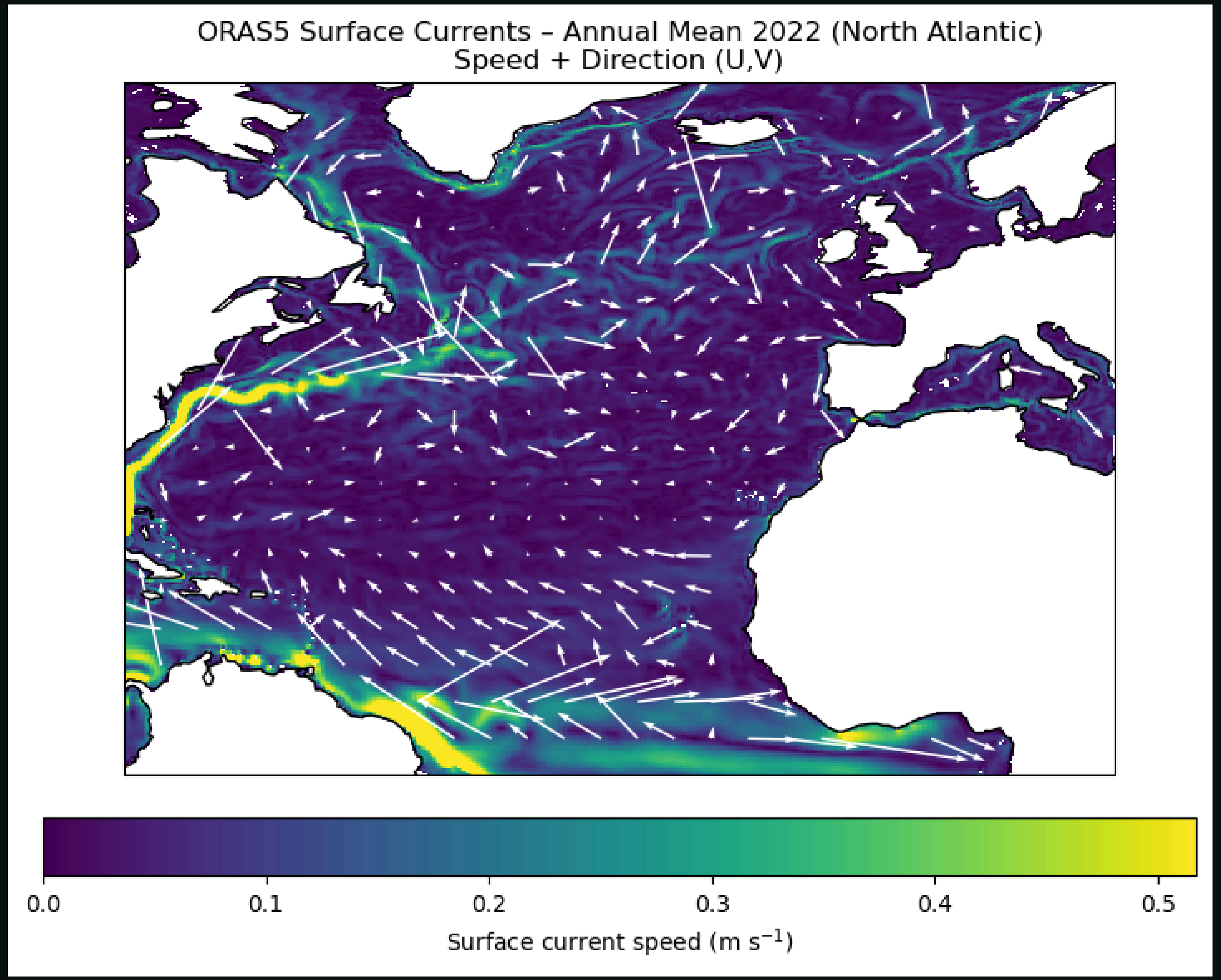
01 ERA5

- Air Temperature
- Precipitation
- Runoff
- All at “single levels”

02 ORAS5

- Sea Surface Temperature (SST)
- Salinity
- Current Velocity
- Heat Flux
- Many variables are 3-dimensional





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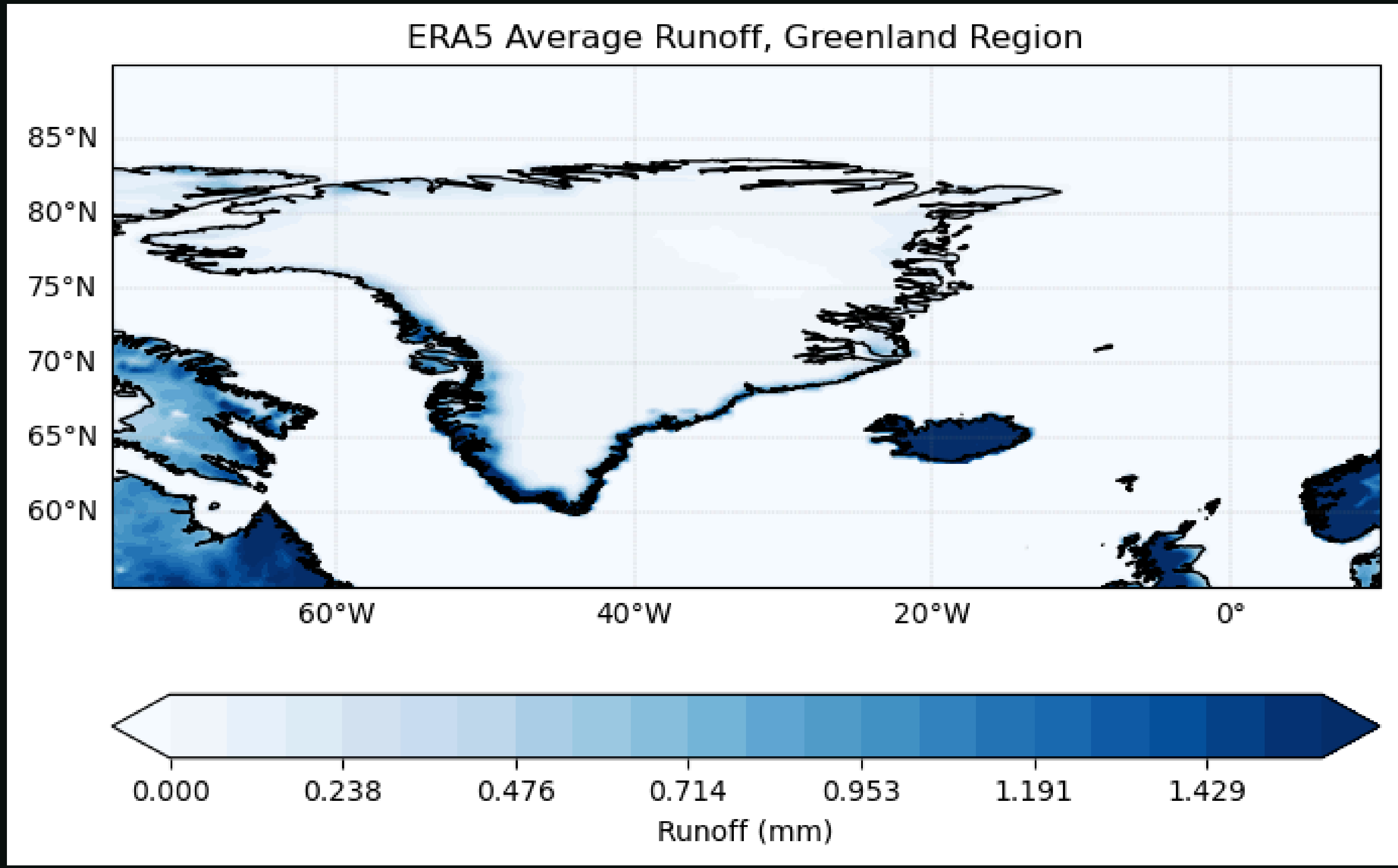
AI + EARLY WARNING SIGNALS



Recent studies suggest that climate systems may exhibit warning signals before major transitions occur. These can include changes in system behaviour and physical warning signals (such as increased freshwater forcing around greenland).

Machine learning methods may help identify these signals across many variables simultaneously.

Possible Physical Indicators include:

- Greenland Freshening
- North Atlantic “Cold Blob”
- Changes in Ocean Heat Transport
- Changes in Freshwater Transport

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CONCLUSIONS

The AMOC is one of the most important components of Earth's climate system and there is growing evidence that it may be weakening. Understanding whether the Atlantic is approaching a critical transition remains a major scientific challenge.

Key points:

- AMOC regulates North Atlantic climate
- Climate change may weaken the circulation
- Past climate records show abrupt changes are possible
- AI provides new tools for climate analysis
- Early detection could improve climate risk assessment